VALIDATION REPORT
CORE INFRASTRUCTURE IQ / OQ

JR Validated Environment

Field	Value
Document ID	JR-VR-001
Document Version	1.0
Execution Date	2026-03-31
IQ/OQ Result	PASS — 6 / 6 tests passed
Status	Final
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-31	Joep Rous	Initial evidence-backed infrastructure validation report (JR-VR-001). Covers TC-CORE-IQ-001 and TC-CORE-OQ-001..005. Results derived from OQ evidence file oq_execution_20260331T215907.txt.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-31
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the automated Installation Qualification (IQ) and Operational Qualification (OQ) tests for the core infrastructure of the JR Validated Environment. It provides objective evidence that the six automated infrastructure test cases from oq/test_core.py were executed and passed under the defined test conditions, meeting the requirements of the Core Validation Plan (JR-VP-001) under 21 CFR 820.70(i) and ISO 13485.

Results are derived directly from the OQ evidence file produced by admin/admin_oq, linking each test case outcome to the raw pytest output.

2. Scope

In scope (automated — this report):

- TC-CORE-IQ-001 — admin_validate passes and produces evidence file
- TC-CORE-OQ-001 — jrc_R_hello.R executes via jrrun
- TC-CORE-OQ-002 — jrc_py_hello.py executes via jrrun
- TC-CORE-OQ-003 — Integrity check detects a tampered tracked file
- TC-CORE-OQ-004 — Bypass protection: R direct invocation fails
- TC-CORE-OQ-005 — Bypass protection: Python direct invocation fails

Manual tests (not automated — must be executed and signed off separately per JR-VP-001):

- OQ-01 — R environment rebuild from scratch (deletes ~/.renv/<PROJECT_ID>)
- OQ-02 — Python environment rebuild from scratch (deletes ~/.venvs/<PROJECT_ID>)
- OQ-06 — Auto-rebuild triggered by missing renv hash
- OQ-07 — admin_install_R --add <package>
- OQ-08 — admin_install_Python --add <package>

Out of scope:

- Community script OQ tests (JR-OQ-001, statistics_validation_report)
- DoE script OQ tests (JR-OQ-001)
- Module-specific tests (JR-VR-CAP-001, JR-VR-MSA-001, etc.)

3. Reference Documents

Document ID	Title	Location
JR-VP-001	Core Infrastructure Validation Plan	docs/core_validation_plan.pdf
JR-OQ-001	Validation Report — Community Script Suite OQ (Core + DoE)	docs/statistics_validation_report.pdf
OQ Evidence	OQ execution evidence file (pytest output + header)	oq_execution_20260331T215907.txt

4. Test Environment

Item	Details
Execution date / time	2026-03-31T21:59:07
Host	Joeps-MacBook-Air.local

Operating system	15.7.3
R version	4.5.3
Python version	3.11.9
pytest version	pytest 8.3.4
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (dedicated OQ environment)
Test runner	admin/admin_oq
Test file	oq/test_core.py
Integrity hash	admin/project_integrity.sha256
OQ evidence file	oq_execution_20260331T215907.txt

5. Test Execution Results

5.1 Summary

Metric	Value
Total test cases executed	6
Passed	6
Failed	0
Errors	0
IQ/OQ Result	**PASS**

5.2 Individual Test Case Results

Results are read directly from the OQ evidence file. The evidence file contains the full pytest output including CMD, OUT (every line of script output), and EXIT for each test invocation.

TC	Type	Description	Acceptance Criteria	Result
TC-CORE-IQ-001	IQ	admin_validate passes and writes evidence file	Exit 0, 'PASSED' in output, timestamped evidence file written to ~/.jrscript/<PROJECT_ID>/validation/.	**PASS**
TC-CORE-OQ-001	OQ	jrc_R_hello.R executes successfully via jrrun	Exit 0, 'Validation' and 'Done' in output.	**PASS**
TC-CORE-OQ-002	OQ	jrc_py_hello.py executes successfully via jrrun	Exit 0 (or timeout after 15 s with GUI window open), 'Validation' in output.	**PASS**
TC-CORE-OQ-003	OQ	Integrity check detects a tampered tracked file	Non-zero exit and integrity failure message after one byte appended to bin/jrrun.	**PASS**
TC-CORE-OQ-004	OQ	Bypass protection — R direct invocation fails	Non-zero exit and 'RENV_PATHS_ROOT' in error output when Rscript called directly.	**PASS**
TC-CORE-OQ-005	OQ	Bypass protection — Python direct invocation fails	Non-zero exit and import error message when system python3 called without the venv.	**PASS**

6. Manual Tests (JR-VP-001)

The following qualification items from JR-VP-001 require deletion of environment directories or interactive admin actions and cannot be automated. They must be executed manually by the installer and signed off before this report is finalised.

OQ Item	Description	How to Execute
OQ-01	R environment rebuild from scratch	Delete ~/.renv/<PROJECT_ID>/. Run jrrun jrc_R_hello.R Validation. Confirm environment rebuilds automatically and script exits 0.

OQ-02	Python environment rebuild from scratch	Delete ~/.venvs/<PROJECT_ID>/. Run jrrun jrc_py_hello.py Validation. Confirm venv rebuilds automatically and script exits 0.
OQ-06	Auto-rebuild triggered by missing renv hash	Delete the renv hash cache entry. Run any R script via jrrun. Confirm renv library is rebuilt and the script runs successfully.
OQ-07	admin_install_R --add <package>	Run admin_install_R --add <testpkg>==<ver>. Confirm package appears in R library and jr_versions output.
OQ-08	admin_install_Python --add <package>	Run admin_install_Python --add <testpkg>==<ver>. Confirm package appears in venv and jr_versions output.

7. Deviations

No deviations from JR-VP-001 were identified for this execution.

8. IQ/OQ Result

Item	Value
IQ/OQ result	**PASS**
Automated test cases	6
Passed	6
Failed	0
Evidence file	oq_execution_20260331T215907.txt
Execution timestamp	2026-03-31T21:59:07

All 6 automated infrastructure test cases passed. Manual tests OQ-01, OQ-02, OQ-06, OQ-07, and OQ-08 must be executed and signed off separately per JR-VP-001 before this report is finalised.

End of Document — JR-VR-001 v1.0